King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 242

Original exam date: 14 April 2025. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If (a, b, c) is the point of intersection between the lines

$$L_1 : x = 1 + 2t,\ y = 1 - t,\ z = 2 - 3t$$

$$L_2 : x = 4 - s,\ y = -1 + 2s,\ z = -4 + 6s$$

then $abc =$

(a) -7

(b) $-\dfrac{20}{9}$

(c) $\dfrac{22}{3}$

(d) $\dfrac{22}{9}$

(e) $\dfrac{11}{9}$

2. The maximum value of the directional derivative of $f(x, y) = xe^{y/x}$ at the point $(1, 2)$ is equal to

(a) $e^2\sqrt{2}$

(b) $2e^2$

(c) e^4

(d) e^2

(e) $e^2 + 1$

3. The set of all points equidistant from the point $(0, 1, 0)$ and the xz-plane is

(a) an elliptic hyperbola

(b) an elliptic paraboloid

(c) a hyperboloid of one sheet

(d) an ellipsoid

(e) a plane

4. If $f(x,y,z) = x^4 - 2xy + 3y^3 + xy^2z^4$, then $f_{xyzz}(x,y,z) =$

(a) $12xz^2$

(b) $6xyz^2$

(c) $12yz^2$

(d) $24yz^2$

(e) $24xyz^2$

5. The domain of the function $f(x,y) = \dfrac{y-1}{\sqrt{x+1}}$ is

(a) $\{(x,y) : x \in [-1,\infty),\ y \in (-\infty,\infty)\}$

(b) $\{(x,y) : x \in (-1,\infty),\ y \in [1,\infty)\}$

(c) $\{(x,y) : x \in (-\infty,\infty),\ y \in (-\infty,\infty)\}$

(d) $\{(x,y) : x \in (1,\infty),\ y \in (-1,\infty)\}$

(e) $\{(x,y) : x \in (-1,\infty),\ y \in (-\infty,\infty)\}$

6. If z is defined implicitly as a differentiable function of x and y by the equation $\sin(2x+y) - 2x\cos(z^2) = 1$, then $\dfrac{\partial z}{\partial x} =$

(a) $\dfrac{\cos(z^2) - \cos(2x+y)}{2xz\sin(z^2)}$

(b) $\dfrac{\cos(2x+y) - \cos(z^2)}{2z\sin(z^2)}$

(c) $\dfrac{\cos(2x+y) - \cos(z^2)}{2xz\cos(z^2)}$

(d) $\dfrac{\cos(2x+y) - 2\cos(z^2)}{xz\sin(z^2)}$

(e) $\dfrac{\cos(z^2) + \cos(2x+y)}{xz\sin(z^2)}$

7. If $z = \sin(x^2y)$, $x = s^2t$, $y = s + \dfrac{1}{t^2}$, then $\dfrac{\partial z}{\partial t} =$

(a) $2s^5t\cos(x^2y)$

(b) $\left(s^3 - \dfrac{s^2}{t}\right)\cos(x^2y)$

(c) $3s^4t^2\cos(x^2y)$

(d) $2st^2\cos(x^2y)$

(e) $\left(2s^5t - \dfrac{s^5}{t}\right)\cos(x^2y)$

8. If $ax + by + cz = 5$ is an equation for the plane through the points $P(2, 2, 1)$ and $Q(-1, 1, -1)$ and is perpendicular to the plane $2x - 3y + z = 3$, then $a + b + c =$

(a) 1

(b) 4

(c) -2

(d) -3

(e) 0

9. An equation for the level curve of $f(x, y) = \ln(x - y)$ that passes through the point $(e^2, 0)$ is given by

(a) $y = x^2$

(b) $y = 2x - 2e^2$

(c) $y = x - e^2$

(d) $y = x - 2$

(e) $y = ex - e^3$

10. If $ax + by + cz = 10$ is an equation for the tangent plane to the surface $x^2 + y^2 + z^2 - 8x - 12y + 4z + 42 = 0$ at the point $(2, 3, -3)$, then $a + b + c =$

(a) 3

(b) 2

(c) 6

(d) 5

(e) 4

11. Let $z = x^2 + y^2$. If (x, y) changes from $(2, 1)$ to $(2.1, 1.05)$, then using differentials, the change in z is approximately equal to

(a) -0.2

(b) 0.5

(c) 0.25

(d) 0.1

(e) 0.03

12. The graph of the equation $z^2 = 1 - 2x^2 - 3y^2$ is

(a) an ellipsoid

(b) a hyperboloid of one sheet

(c) an elliptic paraboloid

(d) an elliptic cone

(e) a hyperboloid of two sheets

13. $\lim_{(x,y)\to(0,0)} \dfrac{y^3}{x^2+y^2} =$

(a) 1

(b) 0

(c) 2

(d) DNE

(e) $\dfrac{1}{2}$

14. If $z = \ln\sqrt{x^2+y^2}$, then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} =$

(a) 1

(b) 0

(c) $\dfrac{z}{2}$

(d) z

(e) $-z$

15. Which one of the following statements is FALSE about the plane $2x - y + z = 4$?

(a) It is parallel to the plane $x - 2y + 2z = 2$

(b) It intersects the x-axis at $(2, 0, 0)$

(c) It has a normal vector $\vec{n} = \langle 2, -1, 1\rangle$

(d) It passes through the point $(2, 1, 1)$

(e) The point $(0, -4, 0)$ lies in the plane

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$t = \frac{4}{3}$, $s = \frac{1}{3}$, point $\left(\frac{11}{3}, -\frac{1}{3}, -2\right)$, $abc = \frac{22}{9}$.
2	**(a)**	$\nabla f(1,2) = \langle -e^2, e^2 \rangle$, magnitude $e^2\sqrt{2}$.
3	**(b)**	$x^2 + (y-1)^2 + z^2 = y^2 \Rightarrow 2y = x^2 + z^2 + 1$: elliptic paraboloid.
4	**(d)**	Reorder: $f_z = 4xy^2z^3$, $f_{zz} = 12xy^2z^2$, then ∂_x, ∂_y: $24yz^2$.
5	**(e)**	Need $x + 1 > 0$; y unrestricted.
6	**(a)**	$-F_x/F_z = -\frac{2\cos(2x+y) - 2\cos z^2}{4xz\sin z^2} = \frac{\cos z^2 - \cos(2x+y)}{2xz\sin z^2}$.
7	**(a)**	$z_t = (2xys^2 - 2x^2t^{-3})\cos(x^2y) = 2s^5t\cos(x^2y)$.
8	**(d)**	$\mathbf{n} = \overrightarrow{PQ} \times \langle 2, -3, 1 \rangle = \langle -7, -1, 11 \rangle$; plane $7x + y - 11z = 5$, sum -3.
9	**(c)**	$c = \ln e^2 = 2$, so $x - y = e^2$, $y = x - e^2$.
10	**(c)**	$\nabla F = \langle -4, -6, -2 \rangle$, plane $2x + 3y + z = 10$, sum 6.
11	**(b)**	$dz = 4(0.1) + 2(0.05) = 0.5$.
12	**(a)**	$2x^2 + 3y^2 + z^2 = 1$: ellipsoid.
13	**(b)**	Polar: $r\sin^3\theta$, squeezed between $\pm r$, limit 0.
14	**(a)**	$z_x = \frac{x}{x^2+y^2}$, $z_y = \frac{y}{x^2+y^2}$, so the sum is 1.
15	**(a)**	Normals $\langle 2, -1, 1 \rangle$ and $\langle 1, -2, 2 \rangle$ are not parallel, so the planes are not parallel.